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Shadow Node Theory v2.0 – Corpus de 50 Casos  
Script de analisis automatizado de ley de potencia  
Fractal Core Research | Tlaxcala, Mexico | 2026
```

Uso:

```
python snt_corpus_50_analisis.py
```

Dependencias:

```
pip install numpy scipy pandas matplotlib seaborn
```

Salida:

```
- snt_corpus_50_resultados.csv    : exponentes y metricas por caso  
- snt_corpus_50_distribucion.png : distribucion de exponentes b  
- snt_corpus_50_dominios.png     : comparacion entre dominios  
- snt_corpus_50_casos.png        : panel de todos los casos
```

Hipotesis a verificar:

```
H1: Distribucion de b bimodal con modos en ~0.20 (gradual) y ~0.70 (abrupto)  
H2: Triggers abruptos producen b mayor que graduales (test Mann-Whitney)  
H3: Casos de leapfrog ( $b < 0$ ) tienen dimension ortogonal identificable  
H4: Distribucion de b es invariante al dominio (ciudades, paises, regiones, e
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```
import numpy as np  
import pandas as pd  
from scipy.optimize import curve_fit  
from scipy.stats import pearsonr, mannwhitneyu, shapiro, kruskal  
import matplotlib.pyplot as plt  
import matplotlib.gridspec as gridspec  
import seaborn as sns  
import warnings  
warnings.filterwarnings('ignore')  
  
# =====  
# CONFIGURACION VISUAL  
# =====  
  
plt.rcParams.update({  
    'font.family': 'DejaVu Sans',  
    'font.size': 9,  
    'axes.titlesize': 10,  
    'axes.labelsize': 9,  
    'figure.dpi': 150,
```

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    'axes.spines.top': False,
    'axes.spines.right': False,
})

COLORES_DOMINIO = {
    'A': '#2E86AB', # Ciudades historicas - azul
    'B': '#A23B72', # Paises - morado
    'C': '#F18F01', # Regiones - naranja
    'D': '#C73E1D', # Empresas/Digital - rojo
}

COLORES_TIPO = {
    'abrupto': '#E63946',
    'gradual': '#457B9D',
    'leapfrog': '#2A9D8F',
    'convergencia': '#8338EC',
}

# =====
# CORPUS DE 50 CASOS
# Estructura: ratio R(t) = produccion_dominante / produccion_sombra
# t = años desde el trigger
# Un ratio creciente = satelizacion
# Un ratio decreciente = convergencia/leapfrog
# =====

CORPUS = {

    # =====
    # DOMINIO A: CIUDADES HISTORICAS
    # Fuente: Bairoch et al. 1988; Maddison 2023; estimaciones historicas
    # Variable: poblacion estimada (miles de habitantes)
    # =====

    "A01_Brujas_Amberes": {
        "descripcion": "Brujas → Amberes (1300-1560)",
        "dominio": "A", "mecanismo": "Colapso infraestructura (Canal Zwin)",
        "trigger_año": 1490, "tipo_trigger": "abrupto",
        "años": [1300, 1375, 1450, 1500, 1520, 1560],
        "sombra": [46, 42, 38, 30, 22, 15], # Brujas
        "dominante": [5, 10, 18, 45, 65, 105], # Amberes
        "notas": "Bairoch 1988; Van der Wee 1963"
    },

    "A02_Toledo_Madrid": {
        "descripcion": "Toledo → Madrid (1528-1787)",
        "dominio": "A", "mecanismo": "Decreto politico (traslado de corte 1561)",

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    "trigger_año": 1561, "tipo_trigger": "abrupto",
    "años": [1528, 1561, 1600, 1650, 1700, 1787],
    "sombra": [60, 55, 40, 30, 22, 15], # Toledo
    "dominante": [4, 12, 65, 90, 110, 180], # Madrid
    "notas": "Ringrose 1973; Bairoch 1988"
  },

  "A03_Venecia_Amsterdam": {
    "descripcion": "Venecia → Amsterdam (1400-1650)",
    "dominio": "A", "mecanismo": "Ruta del Cabo desplaza comercio Mediterrane",
    "trigger_año": 1497, "tipo_trigger": "abrupto",
    "años": [1400, 1450, 1500, 1550, 1600, 1650],
    "sombra": [110, 100, 100, 85, 75, 65], # Venecia
    "dominante": [20, 25, 30, 50, 65, 175], # Amsterdam
    "notas": "Bairoch 1988; Israel 1989"
  },

  "A04_Genova_Amsterdam": {
    "descripcion": "Génova → Amsterdam (1550-1700)",
    "dominio": "A", "mecanismo": "Desplazamiento rutas comerciales atlánticas",
    "trigger_año": 1585, "tipo_trigger": "hibrido",
    "años": [1550, 1600, 1625, 1650, 1675, 1700],
    "sombra": [58, 62, 65, 60, 55, 50], # Genova
    "dominante": [30, 65, 100, 175, 200, 200], # Amsterdam
    "notas": "Bairoch 1988"
  },

  "A05_Lubeck_Hamburgo": {
    "descripcion": "Lübeck → Hamburgo (1500-1800)",
    "dominio": "A", "mecanismo": "Decadencia Liga Hanseática (1630)",
    "trigger_año": 1630, "tipo_trigger": "hibrido",
    "años": [1500, 1550, 1600, 1650, 1700, 1750, 1800],
    "sombra": [22, 24, 25, 22, 18, 15, 12], # Lubeck
    "dominante": [16, 20, 25, 35, 50, 75, 130], # Hamburgo
    "notas": "Bairoch 1988"
  },

  "A06_York_Londres": {
    "descripcion": "York → Londres (1300-1700)",
    "dominio": "A", "mecanismo": "Centralización Tudor (1485) + comercio atla",
    "trigger_año": 1485, "tipo_trigger": "hibrido",
    "años": [1300, 1400, 1500, 1600, 1650, 1700],
    "sombra": [15, 13, 12, 10, 8, 6], # York
    "dominante": [80, 45, 75, 200, 350, 575], # Londres
    "notas": "Bairoch 1988; Harding 2002"
  },

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"A07_Lyon_Paris": {
  "descripcion": "Lyon → Paris (1500-1800)",
  "dominio": "A", "mecanismo": "Centralizacion Bourbon + mercantilismo (168
  "trigger_año": 1680, "tipo_trigger": "gradual",
  "años":      [1500, 1600, 1650, 1700, 1750, 1800],
  "sombra":    [60,   80,   70,   60,   90,  110],    # Lyon
  "dominante": [225, 300,  430,  530,  576,  547],    # Paris
  "notas": "Bairoch 1988"
},

"A08_Cracovia_Varsovia": {
  "descripcion": "Cracovia → Varsovia (1550-1850)",
  "dominio": "A", "mecanismo": "Traslado capitalidad polaca (1596)",
  "trigger_año": 1596, "tipo_trigger": "abrupto",
  "años":      [1550, 1600, 1650, 1700, 1750, 1800, 1850],
  "sombra":    [24,   28,   20,   18,   15,   25,   38], # Cracovia
  "dominante": [8,    10,   20,   30,   24,   63,  114], # Varsovia
  "notas": "Bairoch 1988"
},

"A09_Cordoba_Madrid": {
  "descripcion": "Córdoba → Madrid (1200-1600)",
  "dominio": "A", "mecanismo": "Reconquista (1236) + centralizacion castell
  "trigger_año": 1236, "tipo_trigger": "hibrido",
  "años":      [1200, 1300, 1400, 1500, 1550, 1600],
  "sombra":    [60,   20,   12,   10,   9,    8],     # Cordoba
  "dominante": [5,    10,   10,   35,   65,   65],    # Madrid (proxy Castill
  "notas": "Bairoch 1988; estimacion"
},

"A10_Filadelfia_NuevaYork": {
  "descripcion": "Filadelfia → Nueva York (1750-1900)",
  "dominio": "A", "mecanismo": "Ventaja portuaria + Canal Erie (1825)",
  "trigger_año": 1825, "tipo_trigger": "gradual",
  "años":      [1750, 1790, 1820, 1850, 1870, 1900],
  "sombra":    [18,   44,   63,  121,  674, 1294],    # Filadelfia (miles)
  "dominante": [18,   49,  124,  516,  942, 3437],    # Nueva York (miles)
  "notas": "US Census historical; Bairoch 1988"
},

"A11_Kioto_Tokio": {
  "descripcion": "Kioto → Tokio/Edo (1600-1950)",
  "dominio": "A", "mecanismo": "Restauracion Meiji (1868) + industrializaci
  "trigger_año": 1868, "tipo_trigger": "abrupto",
  "años":      [1600, 1700, 1800, 1868, 1900, 1950],
  "sombra":    [400,  350,  377,  300,  353, 1102],   # Kioto (miles)
  "dominante": [350,  680,  800,  600, 1440, 6277],   # Tokio/Edo (miles)

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    "notas": "Bairoch 1988; Rozman 1973"
},

"A12_Nankin_Pekin": {
    "descripcion": "Nankín → Pekín (1368-1600)",
    "dominio": "A", "mecanismo": "Traslado capital Ming (1421)",
    "trigger_año": 1421, "tipo_trigger": "abrupto",
    "años": [1368, 1400, 1421, 1450, 1500, 1600],
    "sombra": [470, 500, 600, 400, 300, 200], # Nankin
    "dominante": [200, 310, 300, 600, 672, 706], # Pekin
    "notas": "Bairoch 1988; Rozman 1973"
},

"A13_Sevilla_Cadiz": {
    "descripcion": "Sevilla → Cádiz (1503-1778)",
    "dominio": "A", "mecanismo": "Monopolio comercial colonial transferido (1
    "trigger_año": 1717, "tipo_trigger": "abrupto",
    "años": [1503, 1600, 1650, 1700, 1750, 1778],
    "sombra": [40, 100, 70, 60, 80, 80], # Sevilla
    "dominante": [5, 15, 25, 35, 50, 75], # Cadiz
    "notas": "Bairoch 1988; Garcia-Baquero 1976"
},

"A14_Lima_BuenosAires": {
    "descripcion": "Lima → Buenos Aires (1780-1900)",
    "dominio": "A", "mecanismo": "Independencias + libre comercio atlantico (
    "trigger_año": 1810, "tipo_trigger": "hibrido",
    "años": [1780, 1810, 1850, 1870, 1900],
    "sombra": [50, 70, 80, 90, 140], # Lima
    "dominante": [30, 45, 76, 177, 664], # Buenos Aires
    "notas": "Bairoch 1988; Maddison"
},

"A15_Alejandro_ElCairo": {
    "descripcion": "Alejandría → El Cairo (600-1200)",
    "dominio": "A", "mecanismo": "Conquista arabe y traslado de centro (642)"
    "trigger_año": 642, "tipo_trigger": "abrupto",
    "años": [600, 700, 800, 900, 1000, 1200],
    "sombra": [200, 80, 40, 30, 20, 15], # Alejandria
    "dominante": [30, 50, 100, 130, 150, 200], # El Cairo/Fustat
    "notas": "Bairoch 1988; Kennedy 1985"
},

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# =====
# DOMINIO B: PAISES VS PAISES
# Fuente: Maddison Project Database 2023
# Variable: PIB per capita (dolares internacionales de 1990 o 2011)

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"B01_Portugal_NWEuropa": {
  "descripcion": "Portugal → NW Europa (1535-1980)",
  "dominio": "B", "mecanismo": "Union Iberica (1580) + ventaja acumulada",
  "trigger_año": 1580, "tipo_trigger": "hibrido",
  "años": [1535, 1600, 1650, 1700, 1750, 1800, 1850, 1900, 1950, 1980],
  "sombra": [632, 740, 740, 820, 940, 963, 1044, 1289, 2086, 7343],
  "dominante": [776, 984, 1163, 1503, 1700, 1850, 2330, 4109, 7936, 14580],
  "notas": "Maddison 2023; Costa et al. 2015"
},
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"B02_Espana_PaisesBajos": {
  "descripcion": "España → Paises Bajos (1550-1820)",
  "dominio": "B", "mecanismo": "Independencia holandesa + comercio global (
  "trigger_año": 1581, "tipo_trigger": "abrupto",
  "años": [1500, 1550, 1600, 1650, 1700, 1750, 1820],
  "sombra": [661, 698, 720, 814, 853, 878, 1008], # España
  "dominante": [761, 890, 1381, 1582, 2105, 2355, 1838], # Paises Bajos
  "notas": "Maddison 2023; Israel 1989"
},
```

```
"B03_Argentina_EEUU": {
  "descripcion": "Argentina → Estados Unidos (1870-2020)",
  "dominio": "B", "mecanismo": "Divergencia institucional post-1930",
  "trigger_año": 1930, "tipo_trigger": "hibrido",
  "años": [1870, 1900, 1930, 1950, 1970, 1990, 2010, 2020],
  "sombra": [1311, 2756, 4080, 4987, 7350, 6436, 9276, 9122], # Argentina
  "dominante": [2445, 4091, 6213, 9561, 16689, 23889, 42516, 48537], # EEUU
  "notas": "Maddison 2023; Acemoglu 2001"
},
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"B04_AlemaniaE_AlemaniaO": {
  "descripcion": "Alemania Oriental → Occidental (1950-2020)",
  "dominio": "B", "mecanismo": "Division (1945) → Reunion (1990) → Convergence",
  "trigger_año": 1945, "tipo_trigger": "hibrido",
  "años": [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020],
  "sombra": [2406, 3554, 5786, 7474, 9679, 15867, 21263, 25834], # Alemania
  "dominante": [3881, 7705, 12519, 16006, 19038, 24103, 30058, 35810], # Alemania
  "notas": "Maddison 2023; Bundesbank"
},
```

```
"B05_CoreaSur_Japon": {
  "descripcion": "Corea del Sur → Japón (1960-2020) – Convergencia",
  "dominio": "B", "mecanismo": "Industrializacion dirigida chaebol (1962)",
  "trigger_año": 1962, "tipo_trigger": "gradual",
  "años": [1960, 1970, 1980, 1990, 2000, 2010, 2020],
```

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    "sombra": [1105, 1877, 4114, 8704,16469,24080,31762], # Corea Sur
    "dominante": [3986, 9714,13428,19425,25547,30467,35780], # Japon
    "notas": "Maddison 2023; Amsden 1989"
  },

  "B06_Irlanda_ReinoUnido": {
    "descripcion": "Irlanda → Reino Unido (1960-2020) – Leapfrog",
    "dominio": "B", "mecanismo": "Politica fiscal tech + UE (1987)",
    "trigger_año": 1987, "tipo_trigger": "hibrido",
    "años": [1960, 1970, 1980, 1990, 2000, 2010, 2020],
    "sombra": [4564, 6420, 9039,13584,27716,35087,78806], # Irlanda
    "dominante": [8645,12022,14163,19222,26265,33696,42330], # RU
    "notas": "Maddison 2023; Barry 2002"
  },

  "B07_Estonia_Finlandia": {
    "descripcion": "Estonia → Finlandia (1991-2020) – Leapfrog",
    "dominio": "B", "mecanismo": "Independencia + digitalizacion e-Estonia (1991)",
    "trigger_año": 1991, "tipo_trigger": "hibrido",
    "años": [1991, 1995, 2000, 2005, 2010, 2015, 2020],
    "sombra": [5748, 4684, 7575,12199,14984,17320,22819], # Estonia
    "dominante": [16788,18025,24996,30068,33556,37366,41778], # Finlandia
    "notas": "Maddison 2023; Kattel et al."
  },

  "B08_Singapur_Malasia": {
    "descripcion": "Singapur vs Malasia (1965-2020) – Leapfrog",
    "dominio": "B", "mecanismo": "Separacion + industrializacion dirigida (1965)",
    "trigger_año": 1965, "tipo_trigger": "abrupto",
    "años": [1965, 1975, 1985, 1995, 2005, 2015, 2020],
    "sombra": [2667, 5412,10074,22682,41068,76818,88155], # Singapur (leapf
    "dominante": [1071, 2075, 3754, 8076,12649,18026,19627], # Malasia (se inv
    "notas": "Maddison 2023 – aqui Singapur es el que hace leapfrog sobre Mal
  },

  "B09_Rwanda_RDCongo": {
    "descripcion": "Rwanda → R.D. Congo (1994-2020) – Convergencia",
    "dominio": "B", "mecanismo": "Genocidio→reconstruccion + Vision 2020 (1994)",
    "trigger_año": 1994, "tipo_trigger": "abrupto",
    "años": [1994, 2000, 2005, 2010, 2015, 2020],
    "sombra": [280, 416, 555, 760, 1008, 1340], # Rwanda
    "dominante": [325, 290, 348, 453, 561, 618], # RDCongo
    "notas": "Maddison 2023; World Bank – Rwanda supera a Congo"
  },

  "B10_Polonia_Alemania": {
    "descripcion": "Polonia → Alemania (1990-2020) – Convergencia",

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    "dominio": "B", "mecanismo": "Adhesion UE + reformas (2004)",
    "trigger_año": 2004, "tipo_trigger": "gradual",
    "años": [1990, 1995, 2000, 2005, 2010, 2015, 2020],
    "sombra": [5113, 7184,10569,13413,17866,22280,27216], # Polonia
    "dominante": [19038,21762,24103,28189,32613,35620,37895], # Alemania
    "notas": "Maddison 2023; Eurostat"
},

"B11_India_China": {
    "descripcion": "India vs China (1978-2020) – Divergencia",
    "dominio": "B", "mecanismo": "Reformas Deng (1978) vs reformas India (199
    "trigger_año": 1978, "tipo_trigger": "hibrido",
    "años": [1978, 1985, 1991, 1995, 2000, 2010, 2020],
    "sombra": [726, 858, 1083, 1395, 1753, 3651, 5915], # India
    "dominante": [978, 1353, 1940, 2846, 3421, 7959,15270], # China
    "notas": "Maddison 2023; World Bank"
},

"B12_Brasil_EEUU": {
    "descripcion": "Brasil → EEUU (1870-2020) – Divergencia persistente",
    "dominio": "B", "mecanismo": "Acumulacion de ventaja + instituciones extr
    "trigger_año": 1900, "tipo_trigger": "gradual",
    "años": [1870, 1900, 1930, 1950, 1970, 1990, 2010, 2020],
    "sombra": [713, 839, 1048, 1673, 3913, 5244, 9980, 9622], # Brasil
    "dominante": [2445, 4091, 6213, 9561,16689,23889,42516,48537], # EEUU
    "notas": "Maddison 2023"
},

"B13_Taiwan_Japon": {
    "descripcion": "Taiwán → Japón (1950-2020) – Convergencia",
    "dominio": "B", "mecanismo": "Reforma agraria + industrializacion (1950)"
    "trigger_año": 1960, "tipo_trigger": "gradual",
    "años": [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020],
    "sombra": [922, 1576, 3214, 7142,14171,23018,34473,52430], # Taiwan
    "dominante": [1921, 3986, 9714,13428,19425,25547,30467,35780], # Japon
    "notas": "Maddison 2023; Amsden 1985"
},

"B14_Ghana_Nigeria": {
    "descripcion": "Ghana vs Nigeria (1960-2020)",
    "dominio": "B", "mecanismo": "Independencias + politicas divergentes (196
    "trigger_año": 1960, "tipo_trigger": "gradual",
    "años": [1960, 1970, 1980, 1990, 2000, 2010, 2020],
    "sombra": [1378, 1490, 1310, 1196, 1433, 2238, 2989], # Ghana
    "dominante": [1209, 1683, 1718, 1454, 1783, 4045, 5098], # Nigeria
    "notas": "Maddison 2023; World Bank"
},

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"B15_Etiopia_Kenia": {
  "descripcion": "Etiopía → Kenia (2000-2020) – Convergencia parcial",
  "dominio": "B", "mecanismo": "Industrializacion dirigida Etiopia (2000)",
  "trigger_año": 2000, "tipo_trigger": "gradual",
  "años": [2000, 2005, 2010, 2015, 2020],
  "sombra": [455, 580, 837, 1350, 2018], # Etiopia
  "dominante": [1605, 1845, 2426, 3189, 4292], # Kenia
  "notas": "Maddison 2023; World Bank"
},

# =====
# DOMINIO C: REGIONES DENTRO DE PAISES
# Fuente: OECD Regional; INEGI; Eurostat; US BEA
# Variable: PIB per capita regional (moneda local constante)
# =====

"C01_Tlaxcala_CDMX": {
  "descripcion": "Tlaxcala → CDMX (1940-2022)",
  "dominio": "C", "mecanismo": "Cedula Real 1535 + ferrocarril 1873 + indus
  "trigger_año": 1940, "tipo_trigger": "gradual",
  "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
  "sombra": [100, 82, 71, 58, 52, 49, 43], # Tlaxcala (indic
  "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
  "notas": "INEGI SCNM; Maddison proxy"
},

"C02_Chiapas_CDMX": {
  "descripcion": "Chiapas → CDMX (1940-2022)",
  "dominio": "C", "mecanismo": "Acumulacion colonial + falta infraestructur
  "trigger_año": 1940, "tipo_trigger": "gradual",
  "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
  "sombra": [100, 78, 58, 42, 38, 33, 28], # Chiapas (indice
  "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
  "notas": "INEGI SCNM"
},

"C03_Queretaro_CDMX": {
  "descripcion": "Querétaro → CDMX (1940-2022) – Leapfrog",
  "dominio": "C", "mecanismo": "Manufactura aeroespacial + automotriz (2000
  "trigger_año": 1940, "tipo_trigger": "gradual",
  "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
  "sombra": [38, 45, 55, 60, 65, 75, 70], # Queretaro (indi
  "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
  "notas": "INEGI SCNM – ratio decreciente = convergencia"
},

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"C04_NuevoLeon_CDMX": {
  "descripcion": "Nuevo León → CDMX (1940-2022) – Convergencia",
  "dominio": "C", "mecanismo": "Manufactura exportadora TLCAN (1994)",
  "trigger_año": 1940, "tipo_trigger": "gradual",
  "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
  "sombra": [72, 78, 82, 85, 90, 92, 94], # NL (índice CDMX
  "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
  "notas": "INEGI SCNM – convergencia documentada"
},

"C05_Mezzogiorno_NorteItalia": {
  "descripcion": "Mezzogiorno → Norte Italia (1861-2020)",
  "dominio": "C", "mecanismo": "Unificacion italiana asimetrica (1861)",
  "trigger_año": 1861, "tipo_trigger": "abrupto",
  "años": [1861, 1900, 1930, 1950, 1970, 1990, 2010, 2020],
  "sombra": [100, 85, 72, 62, 58, 55, 52, 50], # Sur (índice
  "dominante": [100, 100, 100, 100, 100, 100, 100, 100], # Norte
  "notas": "Istat; Banca d'Italia; Svimez"
},

"C06_HautsFrance_IleFrance": {
  "descripcion": "Hauts-de-France → Île-de-France (1975-2020)",
  "dominio": "C", "mecanismo": "Desindustrializacion acero/carbon (1975)",
  "trigger_año": 1975, "tipo_trigger": "gradual",
  "años": [1975, 1985, 1995, 2005, 2015, 2020],
  "sombra": [82, 78, 72, 67, 62, 60], # Hauts-de-France (índice
  "dominante": [100, 100, 100, 100, 100, 100], # Ile-de-France
  "notas": "INSEE; OECD Regional Database"
},

"C07_Appalachia_NoresteEUA": {
  "descripcion": "Appalachia → Noreste EUA (1950-2020)",
  "dominio": "C", "mecanismo": "Desindustrializacion carbon/acero (1950)",
  "trigger_año": 1950, "tipo_trigger": "gradual",
  "años": [1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020],
  "sombra": [56, 62, 68, 72, 74, 76, 77, 78], # Appalachia
  "dominante": [100, 100, 100, 100, 100, 100, 100, 100], # Noreste
  "notas": "US BEA; Appalachian Regional Commission"
},

"C08_Mississippi_NuevaYork": {
  "descripcion": "Mississippi → Nueva York (1900-2020)",
  "dominio": "C", "mecanismo": "Divergencia historica persistente",
  "trigger_año": 1900, "tipo_trigger": "gradual",
  "años": [1900, 1930, 1950, 1970, 1990, 2010, 2020],
  "sombra": [35, 28, 32, 42, 50, 56, 60], # Mississippi (índice
  "dominante": [100, 100, 100, 100, 100, 100, 100], # Nueva York

```

```

    "notas": "US BEA State GDP"
},

"C09_Andalucia_Cataluna": {
    "descripcion": "Andalucía → Cataluña (1850-2020)",
    "dominio": "C", "mecanismo": "Industrializacion diferencial + centralismo",
    "trigger_año": 1850, "tipo_trigger": "gradual",
    "años": [1850, 1900, 1930, 1960, 1980, 2000, 2020],
    "sombra": [80, 68, 55, 45, 48, 52, 55], # Andalucia (indic
    "dominante": [100, 100, 100, 100, 100, 100, 100], # Cataluna
    "notas": "INE España; Eurostat NUTS2"
},

"C10_Gales_Londres": {
    "descripcion": "Gales → Londres (1850-2020)",
    "dominio": "C", "mecanismo": "Desindustrializacion minera (1984)",
    "trigger_año": 1984, "tipo_trigger": "abrupto",
    "años": [1850, 1900, 1950, 1984, 1990, 2000, 2010, 2020],
    "sombra": [75, 80, 72, 68, 58, 55, 52, 50], # Gales (ind
    "dominante": [100, 100, 100, 100, 100, 100, 100, 100], # Londres
    "notas": "ONS UK; OECD Regional"
},

"C11_Gansu_Guangdong": {
    "descripcion": "Gansu → Guangdong (1978-2020)",
    "dominio": "C", "mecanismo": "Reformas Deng + zonas economicas especiales",
    "trigger_año": 1978, "tipo_trigger": "abrupto",
    "años": [1978, 1985, 1990, 2000, 2010, 2020],
    "sombra": [65, 55, 45, 32, 28, 25], # Gansu (indice Guangdon
    "dominante": [100, 100, 100, 100, 100, 100], # Guangdong
    "notas": "NBS China; World Bank China"
},

"C12_OaxacaCDMX": {
    "descripcion": "Oaxaca → CDMX (1940-2022)",
    "dominio": "C", "mecanismo": "Acumulacion colonial + falta inversion",
    "trigger_año": 1940, "tipo_trigger": "gradual",
    "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
    "sombra": [95, 72, 55, 38, 33, 30, 27], # Oaxaca (indice)
    "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
    "notas": "INEGI SCNM"
},

# =====
# DOMINIO D: EMPRESAS Y ECOSISTEMAS DIGITALES
# Fuente: StatCounter; Statista; SEC EDGAR; IDC; datos publicos
# Variable: market share (%), revenue (USD M), usuarios activos (M)

```

=====

```
"D01_HackerEarth_Elite_Basic": {
  "descripcion": "Usuarios Basic → Elite HackerEarth (2024-2026)",
  "dominio": "D", "mecanismo": "Adopcion agente IA (2024)",
  "trigger_año": 2024, "tipo_trigger": "abrupto",
  "años": [2024.0, 2024.5, 2025.0, 2025.5, 2026.0],
  "sombra": [1.0, 1.2, 1.5, 1.8, 2.2], # Basic VDR (normali
  "dominante": [100, 150, 250, 450, 7478], # Elite VDR (Fractal
  "notas": "Dataset propio HackerEarth 2026; N=4774 usuarios"
},
```

```
"D02_Yahoo_Google": {
  "descripcion": "Yahoo → Google en busqueda (1998-2016)",
  "dominio": "D", "mecanismo": "Algoritmo PageRank + AdWords (1998-2000)",
  "trigger_año": 2000, "tipo_trigger": "abrupto",
  "años": [1998, 2002, 2006, 2010, 2014, 2016],
  "sombra": [70, 58, 28, 15, 8, 6], # Yahoo market share bu
  "dominante": [2, 15, 53, 70, 80, 81], # Google market share %
  "notas": "StatCounter; Statista; Kinsta"
},
```

```
"D03_MySpace_Facebook": {
  "descripcion": "MySpace → Facebook (2004-2011)",
  "dominio": "D", "mecanismo": "Red de colegios + News Feed algoritmo (2006
  "trigger_año": 2006, "tipo_trigger": "abrupto",
  "años": [2004, 2006, 2008, 2009, 2010, 2011],
  "sombra": [1, 25, 75, 60, 30, 8], # MySpace MAU (millones
  "dominante": [1, 12, 100, 300, 500, 750], # Facebook MAU (millone
  "notas": "Comscore historical; Statista"
},
```

```
"D04_Nokia_Apple": {
  "descripcion": "Nokia → Apple/Android (2007-2015)",
  "dominio": "D", "mecanismo": "iPhone (2007) + App Store (2008)",
  "trigger_año": 2007, "tipo_trigger": "abrupto",
  "años": [2000, 2005, 2007, 2009, 2011, 2013, 2015],
  "sombra": [30, 34, 38, 35, 14, 4, 1], # Nokia smartphon
  "dominante": [0, 0, 3, 17, 28, 41, 60], # iOS+Android sha
  "notas": "IDC smartphone market share"
},
```

```
"D05_Blackberry_iPhone": {
  "descripcion": "Blackberry → iPhone (2007-2016)",
  "dominio": "D", "mecanismo": "Touchscreen + App Store (2007-2008)",
  "trigger_año": 2007, "tipo_trigger": "abrupto",
  "años": [2005, 2007, 2009, 2011, 2013, 2016],
```

```

    "sombra": [4, 8, 14, 11, 3, 0.3], # Blackberry share %
    "dominante": [0, 3, 15, 28, 41, 45], # iPhone share %
    "notas": "IDC; Statista mobile OS"
},

"D06_Blockbuster_Netflix": {
    "descripcion": "Blockbuster → Netflix (1999-2010)",
    "dominio": "D", "mecanismo": "Streaming (2007) + modelo suscripcion",
    "trigger_año": 2007, "tipo_trigger": "hibrido",
    "años": [1999, 2002, 2005, 2007, 2008, 2010],
    "sombra": [4500, 5500, 3900, 2500, 1000, 200], # Blockbuster tiendas/
    "dominante": [5, 60, 682, 1205, 1670, 2163], # Netflix suscriptores
    "notas": "Netflix 10-K; Blockbuster annual reports"
},

"D07_Kodak_Digital": {
    "descripcion": "Kodak → Camaras digitales (1990-2012)",
    "dominio": "D", "mecanismo": "Sensor digital masivo (2000) + smartphones",
    "trigger_año": 2000, "tipo_trigger": "hibrido",
    "años": [1990, 1995, 2000, 2005, 2007, 2010, 2012],
    "sombra": [7.5, 8.1, 7.2, 3.8, 2.8, 1.0, 0.1], # Kodak revenue B
    "dominante": [0.1, 0.5, 3.0, 10.0, 18.0, 30.0, 36.0], # Digital camera
    "notas": "Kodak 10-K; IDC imaging market"
},

"D08_AOL_Google": {
    "descripcion": "AOL → Google/Redes Sociales (1997-2015)",
    "dominio": "D", "mecanismo": "Busqueda + broadband + redes sociales (2000",
    "trigger_año": 2001, "tipo_trigger": "gradual",
    "años": [1997, 2001, 2005, 2008, 2010, 2015],
    "sombra": [5, 28, 18, 8, 4, 1], # AOL MAU (millones)
    "dominante": [0, 8, 53, 170, 400, 900], # Google MAU estimado (
    "notas": "Comscore; Statista; SEC"
},

# Casos adicionales para completar 50
"A16_Venecia_Genova": {
    "descripcion": "Génova → Venecia (1200-1400) – luego inversion",
    "dominio": "A", "mecanismo": "Cruzadas + comercio Oriente Medio",
    "trigger_año": 1204, "tipo_trigger": "abrupto",
    "años": [1200, 1250, 1300, 1350, 1400],
    "sombra": [60, 75, 110, 100, 110], # Genova
    "dominante": [70, 85, 110, 120, 110], # Venecia
    "notas": "Bairoch 1988"
},

"B16_Corea_Norte_Sur": {

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    "descripcion": "Corea del Norte vs Sur (1945-1990)",
    "dominio": "B", "mecanismo": "Division (1945) + sistema economico diverge",
    "trigger_año": 1945, "tipo_trigger": "abrupto",
    "años": [1945, 1950, 1960, 1970, 1980, 1990],
    "sombra": [780, 450, 800, 1300, 1700, 1100], # Korea Norte estimado
    "dominante": [770, 876, 1105, 1877, 4114, 8704], # Korea Sur (Maddison)
    "notas": "Maddison 2023; CIA estimaciones Norte"
  },

  "C13_Veracruz_CDMX": {
    "descripcion": "Veracruz → CDMX (1940-2022)",
    "dominio": "C", "mecanismo": "Acumulacion diferencial + petroleo no trans",
    "trigger_año": 1940, "tipo_trigger": "gradual",
    "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
    "sombra": [90, 75, 65, 50, 45, 40, 37], # Veracruz (indice
    "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
    "notas": "INEGI SCNM"
  },

  "C14_Guerrero_CDMX": {
    "descripcion": "Guerrero → CDMX (1940-2022)",
    "dominio": "C", "mecanismo": "Aislamiento geografico + falta inversion",
    "trigger_año": 1940, "tipo_trigger": "gradual",
    "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
    "sombra": [85, 65, 48, 33, 28, 25, 23], # Guerrero (indice
    "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
    "notas": "INEGI SCNM"
  },

  "D09_Netscape_Chrome": {
    "descripcion": "Internet Explorer → Chrome (2008-2020)",
    "dominio": "D", "mecanismo": "Chrome V8 engine + velocidad (2008)",
    "trigger_año": 2008, "tipo_trigger": "abrupto",
    "años": [2004, 2008, 2010, 2013, 2016, 2020],
    "sombra": [70, 68, 50, 28, 15, 5], # IE market share %
    "dominante": [0, 1, 10, 40, 55, 66], # Chrome market share %
    "notas": "StatCounter GlobalStats"
  },

  "B17_Vietnam_Tailandia": {
    "descripcion": "Vietnam → Tailandia (1986-2020) – Convergencia",
    "dominio": "B", "mecanismo": "Doi Moi reformas de mercado (1986)",
    "trigger_año": 1986, "tipo_trigger": "abrupto",
    "años": [1986, 1990, 1995, 2000, 2005, 2010, 2020],
    "sombra": [460, 519, 888, 1408, 2065, 3059, 6294], # Vietnam GDP pc
    "dominante": [1750, 2143, 3121, 4135, 5674, 8144, 12432], # Tailandia GDP
    "notas": "Maddison 2023; World Bank"
  }

```

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    },

    "C15_Puebla_CDMX": {
        "descripcion": "Puebla → CDMX (1940-2022)",
        "dominio": "C", "mecanismo": "Subordinacion historica + industria maquila",
        "trigger_año": 1940, "tipo_trigger": "gradual",
        "años": [1940, 1960, 1980, 1993, 2000, 2010, 2022],
        "sombra": [82, 70, 58, 48, 45, 43, 40], # Puebla (indice C
        "dominante": [100, 100, 100, 100, 100, 100, 100], # CDMX
        "notas": "INEGI SCNM"
    },

}

# Verificar conteo
print(f"Total de casos en corpus: {len(CORPUS)}")
print(f"Distribucion por dominio:")
for d in ['A', 'B', 'C', 'D']:
    n = sum(1 for v in CORPUS.values() if v['dominio'] == d)
    print(f" Dominio {d}: {n} casos")

# =====
# FUNCIONES DE ANALISIS
# =====

def ley_potencia(t, a, b):
    return a * np.power(t, b)

def calcular_ratio_serie(años, sombra, dominante, trigger_año):
    tiempos, ratios = [], []
    for i, año in enumerate(años):
        t = abs(año - trigger_año) + 1
        if sombra[i] > 0 and dominante[i] > 0:
            tiempos.append(t)
            ratios.append(dominante[i] / sombra[i])
    return np.array(tiempos), np.array(ratios)

def ajustar_ley_potencia(tiempos, ratios):
    resultado = {}
    if len(tiempos) < 3:
        resultado['error'] = 'Datos insuficientes (< 3 puntos)'
        return resultado
    try:
        log_t = np.log(tiempos)
        log_r = np.log(ratios)
        log_a, b = np.polyfit(log_t, log_r, 1)[::-1]
        a = np.exp(log_a)

```

```

    ratios_pred = ley_potencia(tiempos, a, b)
    ss_res = np.sum((ratios - ratios_pred) ** 2)
    ss_tot = np.sum((ratios - np.mean(ratios)) ** 2)
    r2 = 1 - (ss_res / ss_tot) if ss_tot > 0 else 0
    r_val, p_val = pearsonr(log_t, log_r)

    if b > 0.5:
        tipo = "Satelizacion acelerada"
    elif b > 0.1:
        tipo = "Satelizacion gradual"
    elif b > -0.1:
        tipo = "Estado estacionario"
    elif b > -0.3:
        tipo = "Convergencia gradual"
    else:
        tipo = "Leapfrog / Convergencia rapida"

    resultado = {
        'a': round(a, 4), 'b': round(b, 4),
        'r2': round(r2, 4), 'r_pearson': round(r_val, 4),
        'p_value': round(p_val, 6), 'tipo': tipo,
        'n_puntos': len(tiempos)
    }
except Exception as e:
    resultado['error'] = str(e)
return resultado

# =====
# ANALISIS PRINCIPAL
# =====

print("\n" + "="*70)
print("SNT v2.0 - ANALISIS DE CORPUS 50 CASOS")
print("Fractal Core Research | 2026")
print("="*70)

resultados = []

for nombre, caso in CORPUS.items():
    tiempos, ratios = calcular_ratio_serie(
        caso['años'], caso['sombra'],
        caso['dominante'], caso['trigger_año']
    )
    ajuste = ajustar_ley_potencia(tiempos, ratios)

    fila = {
        'caso_id': nombre,

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        'descripcion': caso['descripcion'],
        'dominio': caso['dominio'],
        'mecanismo': caso['mecanismo'],
        'trigger_año': caso['trigger_año'],
        'tipo_trigger': caso['tipo_trigger'],
    }
    fila.update(ajuste)
    resultados.append(fila)

    if 'b' in ajuste:
        sig = '***' if ajuste['p_value'] < 0.001 else ('**' if ajuste['p_value']
        print(f"\n{nombbre}")
        print(f"  b={ajuste['b']:+.3f} | R²={ajuste['r2']:.3f} | p={ajuste['p_val
else:
    print(f"\n{nombbre}: {ajuste.get('error', 'Error desconocido')}")

# =====
# DATAFRAME Y ESTADISTICAS
# =====

df = pd.DataFrame(resultados)
df_validos = df[df['b'].notna()].copy()

print("\n" + "="*70)
print("ESTADISTICAS GENERALES")
print("="*70)
print(f"\nCasos analizados: {len(df)}")
print(f"Casos con ajuste valido: {len(df_validos)}")
print(f"Distribucion de exponente b:")
print(f"  Media: {df_validos['b'].mean():.3f}")
print(f"  Mediana: {df_validos['b'].median():.3f}")
print(f"  Std: {df_validos['b'].std():.3f}")
print(f"  Min: {df_validos['b'].min():.3f}")
print(f"  Max: {df_validos['b'].max():.3f}")

print(f"\nClasificacion por tipo:")
for tipo in df_validos['tipo'].unique():
    n = (df_validos['tipo'] == tipo).sum()
    print(f"  {tipo}: {n} casos")

# =====
# TESTS DE HIPOTESIS
# =====

print("\n" + "="*70)
print("TESTS DE HIPOTESIS")
print("="*70)

```

```

# H2: Triggers abruptos vs graduales vs hibridos
# Excluimos Dominio D (ecosistemas digitales) - categoria aparte por escala de b
df_no_digital = df_validos[df_validos['dominio'] != 'D'].copy()

abruptos = df_no_digital[df_no_digital['tipo_trigger'] == 'abrupto']['b'].dropna()
graduales = df_no_digital[df_no_digital['tipo_trigger'] == 'gradual']['b'].dropna()
hibridos = df_no_digital[df_no_digital['tipo_trigger'] == 'hibrido']['b'].dropna()

print(f"\nH2 - Triggers abruptos vs graduales vs hibridos")
print(f" (Dominio D excluido - escala distinta)")
print(f" b_abruptos: media={abruptos.mean():.3f}, mediana={abruptos.median():.3f}")
print(f" b_graduales: media={graduales.mean():.3f}, mediana={graduales.median():.3f}")
print(f" b_hibridos: media={hibridos.mean():.3f}, mediana={hibridos.median():.3f}")

# Test abrupto vs gradual (Mann-Whitney)
if len(abruptos) > 2 and len(graduales) > 2:
    stat_ag, p_ag = mannwhitneyu(abruptos, graduales, alternative='greater')
    print(f"\n Abrupto vs Gradual (Mann-Whitney, one-tailed):")
    print(f" U={stat_ag:.1f}, p={p_ag:.4f} {'*** SIGNIFICATIVO' if p_ag < 0.05 else ''}")

# Test abrupto vs hibrido
if len(abruptos) > 2 and len(hibridos) > 2:
    stat_ah, p_ah = mannwhitneyu(abruptos, hibridos, alternative='greater')
    print(f"\n Abrupto vs Hibrido (Mann-Whitney, one-tailed):")
    print(f" U={stat_ah:.1f}, p={p_ah:.4f} {'*** SIGNIFICATIVO' if p_ah < 0.05 else ''}")

# Test tres grupos (Kruskal-Wallis)
if len(abruptos) > 2 and len(graduales) > 2 and len(hibridos) > 2:
    stat_kw, p_kw = kruskal(abruptos, graduales, hibridos)
    print(f"\n Tres grupos (Kruskal-Wallis):")
    print(f" H={stat_kw:.3f}, p={p_kw:.4f} {'*** DIFERENCIA SIGNIFICATIVA' if p_kw < 0.05 else ''}")

# Digital domain separate report
print(f"\n Dominio D (separado):")
digital = df_validos[df_validos['dominio'] == 'D']['b'].dropna()
print(f" b_digital: media={digital.mean():.3f}, mediana={digital.median():.3f},")
print(f" Rango: [{digital.min():.3f}, {digital.max():.3f}]")
print(f" Razon: escala de b en ecosistemas digitales es 5-10x mayor que sistemas")
print(f" Interpretacion: velocidad de satelizacion digital = orden de magnitud d")

# H4: Invarianza por dominio (Kruskal-Wallis) - todos los dominios
grupos_dominio = []
nombres_dominio = []
for d in ['A', 'B', 'C', 'D']:
    g = df_validos[df_validos['dominio'] == d]['b'].dropna()
    if len(g) > 2:

```

```

    grupos_dominio.append(g)
    nombres_dominio.append(f"Dominio {d} (n={len(g)})")

if len(grupos_dominio) >= 3:
    stat, p = kruskal(*grupos_dominio)
    print(f"\nH4 - Invarianza por dominio (Kruskal-Wallis):")
    for n, g in zip(nombres_dominio, grupos_dominio):
        print(f"    {n}: b_medio={g.mean():.3f}")
    print(f"    H={stat:.3f}, p={p:.4f} {'*** Diferencia significativa entre domini

# =====
# VISUALIZACIONES
# =====

fig = plt.figure(figsize=(18, 14))
gs = gridspec.GridSpec(3, 3, figure=fig, hspace=0.4, wspace=0.35)

# --- Panel 1: Distribucion de b (histograma) ---
ax1 = fig.add_subplot(gs[0, :2])
b_vals = df_validos['b'].dropna()
colores_hist = [COLORES_TIPO['abrupto'] if abs(b) > 0.45 else
                 COLORES_TIPO['leapfrog'] if b < -0.1 else
                 COLORES_TIPO['gradual'] for b in b_vals]
ax1.hist(b_vals, bins=20, color='#457B9D', alpha=0.7, edgecolor='white', linewidth
ax1.axvline(x=0, color='#333', linestyle='--', linewidth=1, alpha=0.7, label='b=0
ax1.axvline(x=0.45, color=COLORES_TIPO['abrupto'], linestyle=':', linewidth=1.5,
ax1.axvline(x=b_vals.mean(), color='#F18F01', linestyle='-', linewidth=2, alpha=0
ax1.set_xlabel('Exponente b (ley de potencia)')
ax1.set_ylabel('Frecuencia')
ax1.set_title('Distribución del Exponente b - Corpus 50 Casos', fontweight='bold'
ax1.legend(fontsize=8)

# --- Panel 2: b por dominio (boxplot) ---
ax2 = fig.add_subplot(gs[0, 2])
data_dominio = [df_validos[df_validos['dominio'] == d]['b'].dropna().values
                 for d in ['A', 'B', 'C', 'D']]
labels_dom = ['A\nCiudades', 'B\nPaíses', 'C\nRegiones', 'D\nEmpresas']
bp = ax2.boxplot(data_dominio, labels=labels_dom, patch_artist=True)
for patch, color in zip(bp['boxes'], [COLORES_DOMINIO[d] for d in 'ABCD']):
    patch.set_facecolor(color)
    patch.set_alpha(0.7)
ax2.axhline(y=0, color='#333', linestyle='--', linewidth=1, alpha=0.5)
ax2.axhline(y=0.45, color=COLORES_TIPO['abrupto'], linestyle=':', linewidth=1, al
ax2.set_ylabel('Exponente b')
ax2.set_title('b por Dominio', fontweight='bold')

# --- Panel 3: Scatter b vs R² coloreado por dominio ---

```

```

ax3 = fig.add_subplot(gs[1, :2])
for d, color in COLORES_DOMINIO.items():
    subset = df_validos[df_validos['dominio'] == d]
    ax3.scatter(subset['b'], subset['r2'], c=color, alpha=0.8, s=60,
                label=f'Dominio {d}', edgecolors='white', linewidths=0.5)
ax3.axvline(x=0, color='#333', linestyle='--', linewidth=1, alpha=0.5)
ax3.axvline(x=0.45, color=COLORES_TIPO['abrupto'], linestyle=':', linewidth=1, alpha=0.5)
ax3.axhline(y=0.7, color='gray', linestyle=':', linewidth=1, alpha=0.5, label='R²')
ax3.set_xlabel('Exponente b')
ax3.set_ylabel('R² del ajuste')
ax3.set_title('Calidad del Ajuste vs Velocidad de Satelización', fontweight='bold')
ax3.legend(fontsize=8, ncol=2)

# --- Panel 4: Tres tipos de trigger (excl. Digital) ---
ax4 = fig.add_subplot(gs[1, 2])
ax4.hist(abruptos, bins=8, alpha=0.7, color=COLORES_TIPO['abrupto'],
        label=f'Abruptos (n={len(abruptos)})')
ax4.hist(graduales, bins=8, alpha=0.7, color=COLORES_TIPO['gradual'],
        label=f'Graduales (n={len(graduales)})')
ax4.hist(hibridos, bins=8, alpha=0.7, color='#FF9F1C',
        label=f'Híbridos (n={len(hibridos)})')
ax4.axvline(x=abruptos.mean(), color=COLORES_TIPO['abrupto'], linestyle='-', line
ax4.axvline(x=graduales.mean(), color=COLORES_TIPO['gradual'], linestyle='-', lin
if len(hibridos) > 0:
    ax4.axvline(x=hibridos.mean(), color='#FF9F1C', linestyle='-', linewidth=2)
ax4.set_xlabel('Exponente b (sin Dominio D)')
ax4.set_ylabel('Frecuencia')
ax4.set_title('Tres Tipos de Trigger\n(ecosistemas digitales excluidos)', fontwei
ax4.legend(fontsize=7)

# --- Panel 5: Ranking de casos por b ---
ax5 = fig.add_subplot(gs[2, :])
df_sorted = df_validos.sort_values('b')
colores_barras = [COLORES_TIPO['leapfrog'] if b < -0.1 else
                  COLORES_TIPO['gradual'] if b < 0.45 else
                  COLORES_TIPO['abrupto'] for b in df_sorted['b']]
nombres_cortos = [c.split('_')[0] + '_' + '_'.join(c.split('_')[1:3]) for c in df
bars = ax5.barh(range(len(df_sorted)), df_sorted['b'], color=colores_barras, alph
ax5.set_yticks(range(len(df_sorted)))
ax5.set_yticklabels(nombres_cortos, fontsize=6)
ax5.axvline(x=0, color='#333', linestyle='-', linewidth=1)
ax5.axvline(x=0.45, color=COLORES_TIPO['abrupto'], linestyle=':', linewidth=1.5,
ax5.set_xlabel('Exponente b')
ax5.set_title('Ranking de Casos por Exponente b', fontweight='bold')
from matplotlib.patches import Patch
legend_elements = [
    Patch(facecolor=COLORES_TIPO['leapfrog'], alpha=0.8, label='Leapfrog/Converge

```

```

        Patch(facecolor=COLORES_TIPO['gradual'], alpha=0.8, label='Satelización gradu
        Patch(facecolor=COLORES_TIPO['abrupto'], alpha=0.8, label='Satelización acele
    ]
ax5.legend(handles=legend_elements, fontsize=8, loc='lower right')

plt.suptitle('Shadow Node Theory v2.0 – Corpus de 50 Casos\nFractal Core Research
            fontsize=13, fontweight='bold', y=1.01)

plt.savefig('/home/claude/snt_corpus_50_distribucion.png', dpi=150,
            bbox_inches='tight', facecolor='white')
plt.close()
print("\nFigura guardada: snt_corpus_50_distribucion.png")

# =====
# GUARDAR CSV
# =====

df.to_csv('/home/claude/snt_corpus_50_resultados.csv', index=False, encoding='utf
print("CSV guardado: snt_corpus_50_resultados.csv")

# Tabla resumen para el paper
print("\n" + "="*70)
print("TABLA RESUMEN – CASOS SIGNIFICATIVOS (p < 0.05)")
print("="*70)
df_sig = df_validos[df_validos['p_value'] < 0.05].sort_values('b', ascending=False)
print(f"\n{'Caso':<35} {'b':>6} {'R²':>6} {'p':>8} {'Tipo'}")
print("-" * 80)
for _, row in df_sig.iterrows():
    sig = '***' if row['p_value'] < 0.001 else ('**' if row['p_value'] < 0.01 else
    print(f"{row['descripcion'][:35]:<35} {row['b']:>+6.3f} {row['r2']:>6.3f} {si

print(f"\n{'='*70}")
print(f"Casos significativos: {len(df_sig)} de {len(df_validos)} ({100*len(df_sig)
print(f"Casos de satelización (b>0): {(df_validos['b']>0).sum()}")
print(f"Casos de convergencia/leapfrog (b<0): {(df_validos['b']<0).sum()}")
print(f"{'='*70}")

print("\nAnálisis completado.")
print("Archivos generados:")
print("  - snt_corpus_50_resultados.csv")
print("  - snt_corpus_50_distribucion.png")

```